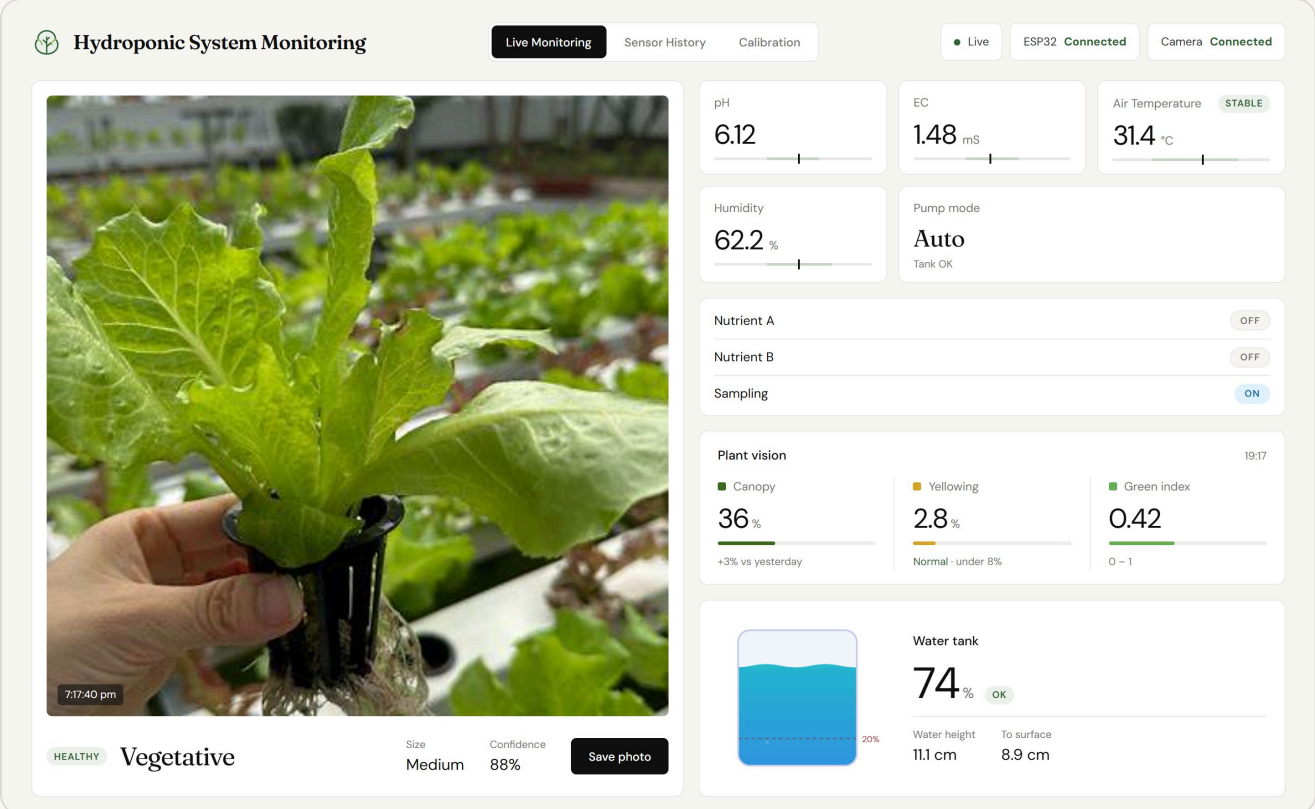
01

Indoor Vertical Garden Monitoring & Automation System

Introduction to a hydroponic lettuce system that measures its own water and air, doses nutrients by itself, watches the plants with a camera, and shows everything on an offline dashboard.



Live Monitoring tab of the dashboard (screenshot taken 13 Nov 2026 with simulated sensor data).

TEAM	CROP	UPDATED
UniKL MFI Robotic Society	Lettuce (*Lactuca sativa*)	13 November 2026

1 · BACKGROUND

Why this project

Hydroponic lettuce grows fast and clean, but only if the nutrient water stays in a narrow range. Checking it by hand every day is slow and easy to forget. This project lets the system check and correct itself.

Problem statement

- **Nutrient strength drifts.** Plants take up nutrients and water at different rates, so EC (nutrient strength) and pH move every day. Lettuce does best at **EC 1.2–1.8 mS/cm** and **pH 5.5–6.5**.
- **Manual checks are infrequent.** A handheld meter gives one reading a day; a problem that starts at night is found the next morning.
- **Pumps can run dry.** If the tank empties while a pump is running, the pump is damaged.
- **Plant health is judged by eye.** Yellowing leaves and slow growth are noticed late and are not recorded.
- **No internet in the grow room.** Cloud dashboards are not an option; everything must work offline.

Objectives

#	Objective	How it is met
1	Measure air temperature, humidity, EC, pH and water level continuously.	ESP32 with DHT22, EC, pH and ultrasonic sensors, one reading every 2 s
2	Keep EC and pH inside the lettuce target range automatically.	Pump A + Pump B auto control in the firmware
3	Protect the pumps from running dry.	Tank-low stop below 20 % water level
4	Record every reading for later analysis.	CSV session logs on the Raspberry Pi 5
5	Check plant health and growth stage with a camera.	MaixCAM photos, leaf-colour check and an AI growth-stage model
6	Show live values, history and calibration on a dashboard, fully offline.	MQTT (Mosquitto) + web dashboard on port 8000, HDMI kiosk

Scope

In scope

- One hydroponic rack with one nutrient tank
- Sensing, nutrient dosing, dry-run protection
- Data logging, camera vision, local dashboard
- Calibration of EC, pH and tank level from the dashboard

Not (yet) in scope

- Heating or cooling (air temperature is only reported)
- Water temperature compensation for EC (DS18B20 planned)
- Grow-light control (planned, via relay)
- Remote access over the internet

2 · SYSTEM AT A GLANCE

What the system is made of

An ESP32 reads the sensors and drives the pumps. A Raspberry Pi 5 logs the data, runs the camera AI and serves the dashboard. They talk over a USB cable and a local MQTT broker. No Wi-Fi or internet is needed at runtime.

Component	Role
ESP32 Dev Module	Reads all sensors, runs pump auto control and tank-low protection, sends one JSON line every 2 s over USB serial (115200 baud).
DHT22	Air temperature and humidity (status band 25–36 °C).
EC sensor (EC V2.1)	Nutrient strength in mS/cm, two-point calibration (1413 µS/cm, 12.88 mS/cm).
pH sensor (PH-4502C)	Acidity of the nutrient water, two-point calibration (pH 7, pH 4).
HC-SR04	Water level in the tank (cm and %), mounted above the water facing down.
2 × 2-channel relay	Pump A (Nutrient A / Pump In), Pump B (Nutrient B / Pump Out), sampling pump (Pump N).
Raspberry Pi 5	Serial → MQTT bridge, CSV logs, history API, camera vision, dashboard, HDMI kiosk.
Sipeed MaixCAM	Plant camera, connected to the Pi by USB-C as a network adapter; serves a photo every 10 s.

Key features

Automatic nutrient dosing

Pump A and Pump B run until EC is 1.2–1.8 mS/cm and pH is 5.5–6.5, then stop. Start delay 60 s after boot.

Dry-run protection

All pumps are forced off when the water level is below 20 % (3 low readings in a row), in auto and manual mode.

Plant vision + AI

Leaf-colour health check and a MobileNetV3 growth-stage model (86 % test accuracy, 10 ms per photo on the Pi 5).

Offline dashboard

Live tiles, 1 h / 24 h / 7 d history, CSV export and guided calibration. Shown full screen on the HDMI monitor at boot.

Project status

Status	Item
DONE	ESP32 firmware (DHT22, EC, pH, HC-SR04), pump auto control, tank-low stop, relay test
DONE	USB serial → MQTT bridge, CSV session logs, history API, dashboard (3 tabs), HDMI kiosk
DONE	MaixCAM camera, leaf-colour health check, 88-photo dataset, growth-stage model
NEXT	More own-camera photos and retrain · MQTT password · DS18B20 water temperature · grow-light relay · RTC battery

> The other documents in this set: **02** ESP32 Pin Configuration · **03** Dataset · **04** System Diagram & Flowchart · **05** Dashboard Overview. The full technical guide is `docs/pdf/Hydro_System_Guide.pdf`.